

STATCHECK

Tier 1 :: Research Methods Errors

At a glance: statcheck automatically extracts reported test statistics from the text of psychology and related research papers — t-tests, F-tests, chi-squared tests, z-tests, and correlations — and recalculates their associated p-values from scratch. It then compares the recalculated p-value to the one reported in the paper, flagging inconsistencies. Errors that cross the $p < .05$ significance threshold are flagged as decision errors — cases where the reported conclusion may be incorrect. No raw data is needed; statcheck works from a machine-readable PDF or text file.

1. Overview

When researchers report the results of statistical tests in the text of a paper, they typically report three numbers: the test statistic, the degrees of freedom, and the p-value. These three numbers are mathematically linked — given any two, the third can be calculated. This means every reported p-value can be independently verified from the test statistic and degrees of freedom alone, without access to the original data.

statcheck, developed by Michèle Nuijten and colleagues at Tilburg University and first released in 2014, automates this verification at scale. It scans a paper's text using pattern-matching rules that recognize standard reporting formats (e.g., $t(38) = 2.31, p = .026$), extracts the test type, statistic, and degrees of freedom, recalculates the p-value using the appropriate distribution function, and checks whether the reported p-value is consistent with the recalculated one.

The tool was designed for the specific reporting conventions of the American Psychological Association (APA) style, which require test statistics to be reported in a standardized format. This constraint is simultaneously its greatest strength — it enables fully automated checking — and its key limitation: it only catches errors in statistics that follow the recognized format, and misses errors in non-standard reporting or in statistics that are not in the extracted set.

statcheck does not detect fabrication or intentional manipulation. A researcher who invents a test statistic and a consistent p-value will pass statcheck without error. What it detects is inconsistency — cases where the reported p-value does not match the test statistic and degrees of freedom, whether due to rounding, transcription error, copy-paste mistakes, or the more concerning possibility of post-hoc adjustment of reported values.

2. Methodological Overview

Extraction via regular expressions

statcheck uses a set of regular expressions — pattern-matching rules applied to paper text — to identify and extract reported test results. The recognized formats cover the most common APA-style test statistics:

t-test: $t(df) = \text{value}, p = \text{value}$

Example: $t(48) = 2.14, p = .037 \rightarrow$ statcheck extracts: type=t, df=48, stat=2.14, p_reported=.037

F-test: $F(df1, df2) = \text{value}, p = \text{value}$

Example: $F(2, 87) = 4.32, p = .016 \rightarrow$ extracts: type=F, df1=2, df2=87, stat=4.32, p_reported=.016

Chi-squared: $\chi^2(df) = \text{value}, p = \text{value}$

Also accepts $r(df) = \text{value}, p = \text{value}$ for correlations and $z = \text{value}, p = \text{value}$ for z-tests.

Recalculation and comparison

For each extracted result, statcheck recalculates the p-value using the standard distribution function in R for that test type: `pt()` for t-statistics, `pf()` for F-statistics, `pchisq()` for chi-squared, `pnorm()` for z-scores. The recalculated p-value is then compared to the reported p-value.

Two thresholds are applied:

Inconsistency	The recalculated p-value differs from the reported p-value by more than would be expected from rounding alone (default threshold: the reported p-value is
---------------	---

	inconsistent if it deviates from the recalculated value at the second decimal place or beyond). Roughly half of detected inconsistencies are benign rounding errors.
Decision error	The recalculated p-value crosses the $p = .05$ significance threshold in the opposite direction from the reported p-value. For example: reported $p = .049$ (significant) but recalculated $p = .051$ (non-significant), or vice versa. Decision errors are the subset of inconsistencies that would change the reported statistical conclusion. These are the primary focus of integrity analysis.

The two-tier classification is important: most inconsistencies are not decision errors. A reported $p = .023$ recalculating to $p = .024$ is an inconsistency but not a decision error. Only inconsistencies where the significance threshold is crossed in the wrong direction warrant concern about whether the reported conclusion is correct.

3. Example Application: Psychology Literature Survey

Nuijten et al. (2016) applied statcheck to a corpus of 30,717 articles published in eight major psychology journals between 1985 and 2013. The analysis extracted and checked 258,050 null hypothesis significance tests. The results were striking:

- Approximately 49% of papers contained at least one reporting inconsistency — a recalculated p-value that did not match the reported one.
- Approximately 13% of papers contained at least one decision error — an inconsistency where the significance conclusion would change.
- Decision errors were not random: they were significantly more likely to favor the researcher's hypothesis (a non-significant result reported as significant) than to work against it. This asymmetry is consistent with selective reporting or post-hoc adjustment, not random transcription error.
- Error rates decreased slightly in recent years but remained substantial throughout the study period. Journals with more rigorous statistical reporting policies had modestly lower error rates.

The paper triggered significant discussion about the reliability of psychological research reporting and prompted several journals to implement statcheck screening at the submission stage. Some journals now run statcheck automatically on all submitted manuscripts and require authors to explain any flagged inconsistencies before peer review.

A follow-up study by Epskamp and Nuijten (2016) applied statcheck to the Open Science Collaboration's Reproducibility Project dataset, finding that decision errors were significantly more common in studies that failed to replicate than in those that successfully replicated — consistent with the hypothesis that some decision errors reflect meaningful manipulation of reported statistics rather than innocent rounding.

4. Requirements & Applicability

statcheck is notable for its low barrier to entry — it can be applied to any paper that reports statistics in a recognizable format, with no access to raw data required.

Machine-readable text	statcheck requires the paper's text in a format that can be parsed by R. PDF files can be processed directly by statcheck via PDF-to-text conversion (the tool handles this internally). Scanned image PDFs — common in older journal archives — cannot be processed and must be OCR-converted first.
APA-style reporting format	statcheck recognizes statistics reported in APA format: test statistic, degrees of freedom in parentheses, equals sign, value, p-value. Non-standard formats (e.g., 'the result was significant at $p < .05$ ' without reporting the test statistic) are not extracted and not checked. Articles in fields that do not follow APA reporting conventions will have low extraction rates.
Test types supported	t, F, r (correlation), χ^2 (chi-squared), and z. Results from regression coefficients, Bayesian analyses, multilevel models, structural equation models, mixed-effects models, and most non-parametric tests are not extracted unless they are expressed in terms of a supported test statistic.

Raw data required?	No. statcheck operates entirely on the text of the published paper. This is its key practical advantage and the reason it can be applied at scale to large literature corpora.
Language	The pattern-matching rules are designed for English-language papers. Non-English papers or papers with unconventional typographic conventions (e.g., commas as decimal separators in some European journals) may have lower extraction rates or higher false-positive rates.
One-tailed vs. two-tailed	statcheck assumes two-tailed tests by default. One-tailed p-values that are reported as such will appear as inconsistencies because they are half the two-tailed value. The tool provides an option to check one-tailed results explicitly, but this must be applied carefully.

5. Error Taxonomy: What statcheck Finds and What It Misses

Understanding statcheck's output requires distinguishing between the types of errors it can and cannot detect. This matters for interpreting a clean statcheck result — it is not a certification of accuracy, only of consistency in the checked statistics.

What statcheck can detect

- Transcription errors: a test statistic, degrees of freedom, or p-value that was mistyped when the paper was written or typeset.
- Copy-paste errors: statistics copied from statistical software output but with a value from the wrong cell or row.
- Rounding errors: a p-value rounded to fewer decimal places than the test statistic justifies, causing a reported $p = .05$ for a test that actually gives $p = .054$.
- Decision errors: the subset of inconsistencies large enough to change the significance conclusion. These may reflect any of the above causes, or may reflect more deliberate adjustment of a marginal result.

What statcheck cannot detect

- Fabricated but internally consistent statistics: a researcher who invents a t-statistic of 2.31, $df = 38$, and correctly calculates $p = .025$ will produce no inconsistency. statcheck verifies the math, not the underlying data.
- Errors in non-extracted statistics: only test results in recognized formats are checked. A paper may have dozens of unextracted statistics that contain errors.
- Wrong test applied: a researcher who applies a paired t-test to independent samples, or uses the wrong degrees of freedom for an ANOVA, will produce internally consistent statistics that are nonetheless incorrect. statcheck has no way to assess whether the right test was used.
- Effect size errors: Cohen's d, eta-squared, and other effect sizes are not extracted or checked by statcheck.
- Selective reporting: statcheck cannot tell whether the extracted statistics represent all tests conducted or only the subset that was significant. A paper that reports five significant results from fifty conducted analyses will pass statcheck cleanly on all five reported tests.

6. Limitations

- The false positive rate for decision errors is non-trivial. Nuijten et al. estimated that approximately 1.6% of all extracted results are decision errors — but in any individual paper with a small number of results, even a single flagged decision error should be interpreted cautiously. Rounding and transcription error are far more common than deliberate manipulation.
- Extraction rate varies substantially across papers. Papers with many statistics reported in tables (rather than text) or in non-standard formats will have low extraction rates. A paper with zero extracted results cannot be evaluated; this is not evidence of accuracy.
- statcheck does not evaluate the appropriateness of the statistical test used, the validity of the degrees of freedom, or whether the reported test matches the one described in the methods section. These require human review.

- The tool's focus on p-values is a limitation in an era of increasing effect size and confidence interval reporting. As journals shift to requiring confidence intervals and discouraging binary p-value reporting, statcheck's coverage of reported results will decrease.
- Checking at scale requires careful interpretation of aggregate results. A 13% decision error rate across a literature does not mean 13% of papers contain deliberate falsification — the vast majority reflect mundane errors. Individual paper-level results should never be publicly attributed to misconduct without further investigation.

7. How to Apply

Web interface (statcheck.io)

The quickest way to check a single paper is to paste its text into the statcheck web app at statcheck.io. The app returns a table of extracted statistics with recalculated p-values, inconsistency flags, and decision error flags. No installation required.

R package

- Install: `install.packages('statcheck')` in R.
- Check a single PDF: `result <- statcheck('path/to/paper.pdf')`. Returns a data frame with one row per extracted statistic.
- Check a folder of PDFs: `result <- statcheckPDFFolder('path/to/folder/')`. Scales to hundreds of papers automatically.
- Check pasted text: `result <- statcheck('t(38) = 2.14, p = .037', text = TRUE)`. Useful for testing specific results without processing a full PDF.
- Inspect results: result shows columns for Source, Statistic, df1, df2, Test value, p_reported, p_computed, Reported_conclusion, Correct, Error, and Decision_error. Filter on `Decision_error == TRUE` for the most consequential flags.

Interpreting output

A clean statcheck output (no inconsistencies) is not a certification of accuracy — it means all extracted statistics are internally consistent. A flagged decision error warrants re-examination of the original analysis and, where possible, re-running the test with the reported data. Authors should be contacted for clarification before any public allegation of error is made.

8. Related Tools

GRIM Test	Checks whether reported means are arithmetically possible given integer response data and sample size. Complementary to statcheck: GRIM targets means while statcheck targets test statistics. Both operate on published summary statistics without raw data.
SPRITE	Reconstructs possible underlying datasets from reported M, SD, and N. Deeper than statcheck but requires integer-scale data. Together, GRIM, SPRITE, and statcheck form a comprehensive Tier 1 screening toolkit for published results.
p-curve	Uses the distribution of significant p-values to assess evidential value across a literature. Where statcheck flags individual inconsistent p-values within papers, p-curve assesses whether the pattern of significant p-values across papers reflects genuine effects or selective reporting.
Registered Reports	The structural prevention: if study outcomes are committed to in advance, the post-hoc adjustment of p-values that statcheck detects becomes impossible. Journals requiring Registered Reports show substantially lower decision error rates in statcheck analyses.

Key Citations

Nuijten, M. B., Hartgerink, C. H. J., van Assen, M. A. L. M., Epskamp, S., & Wicherts, J. M. (2016). The prevalence of statistical reporting errors in psychology (1985–2013). *Behavior Research Methods*, 48, 1205–1226. <https://doi.org/10.3758/s13428-015-0664-2>

- Epskamp, S. & Nuijten, M. B. (2016). statcheck: Extract statistics from articles and recompute p-values (R package version 1.2.2). <https://cran.r-project.org/package=statcheck>
- Nuijten, M. B. (2018). Practical tools and strategies for researchers to increase replicability. *Developmental Medicine & Child Neurology*, 61(5), 535–539. <https://doi.org/10.1111/dmcn.14054>
- Epskamp, S. & Nuijten, M. B. (2016). Statcheck results of papers in OSC reproducibility project. OSF. <https://osf.io/a75p2/>
- Bakker, M. & Wicherts, J. M. (2011). The (mis)reporting of statistical results in psychology journals. *Behavior Research Methods*, 43(3), 666–678. <https://doi.org/10.3758/s13428-011-0089-5>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. <https://doi.org/10.1126/science.aac4716>
- John, L. K., Loewenstein, G., & Prelec, D. (2012). Measuring the prevalence of questionable research practices with incentives for truth telling. *Psychological Science*, 23(5), 524–532. <https://doi.org/10.1177/0956797611430953>